DATA ANALYST PORTFOLIO PROJECT

San Francisco Restaurant Hygiene & Safety Analytics

Data-Driven Public Health Surveillance & Risk Mitigation Strategy (2014 - 2016)

Author: Gilang Anugrah | Tools: Python, SQL, DAX, Power BI | Client: SF Department of Public Health (SFDPH)

CONTEXT & BUSINESS OVERVIEW

Executive Summary: Transforming Inspection Data into Action

TOTAL INSPECTIONS

25,834

Across 6,078 unique restaurants

AVG HYGIENE SCORE

91.3 / 100

Citywide standard: Good / Excellent

TOTAL VIOLATIONS

40,673

86% routine maintenance non-compliances

HIGH-RISK EXPOSURE

13.8%

Critical public health hazards (Vermin/Temp)

Business Context & Mission

- Stakeholder: San Francisco Department of Public Health (SFDPH).
- The Challenge: SFDPH oversees 6,000+ commercial food establishments with finite inspector headcounts and tight municipal budget constraints.
- The Core Problem: Prioritizing routine inspections reactively leads to blind spots in severe contamination prevention.
- The Mission: Harness 53,000+ historical records to identify high-risk hotspots, seasonal pest outbreaks, and optimize patrol scheduling.

Key Analytical Discoveries

- Citywide Baseline: 68% of food venues operate at Excellent/Good scores (>=90). Less than 2% fall into severe non-compliance (<70).
- Geographic Disparity: Commercial downtown corridors (Financial District 94104/94111) suffer from 20-23% high-risk rates, vs residential suburbs (~10-12%).
- Summer Seasonality: Vermin & pest infractions surge by +85% in May-June compared to winter months, establishing a clear timing for preventive intervention.
- Enforcement ROI: Mandatory re-inspections for failing venues yielded an average +8.57 score rebound, confirming enforcement efficacy.

METHODOLOGY & TECH STACK

Analytics Architecture: CRISP-DM Framework & Toolchain

1. Data Cleaning (Python)

Handled grain duplication, corrected (0,0) GPS anomalies, standard 5-digit zip parsing, removed 50 exact duplicates, exported clean CSVs.

Pandas, NumPy, KaggleHub

2. Database & SQL EDA

Loaded data into SQLite & T-SQL (SSMS). Implemented CTEs, Window Functions (LAG, DENSE_RANK, SUM OVER) for Pareto & seasonality analysis.

SQLite, SSMS (T-SQL)

3. DAX & BI Modeling

Constructed Star-Schema style summary data model, authored dynamic DAX measures (% High Risk Rate, Compliance Tiers), professional custom palette.

Power BI Desktop, DAX

4. Decision Delivery

Translated diagnostic metrics into executive insights, formulated seasonal prevention protocols, and delivered high-impact interactive visuals.

Executive Dashboard, PDF/PPTX

FINDING 1: SCORE DISTRIBUTION

Citywide Compliance: 68% Operate in Good/Excellent Standing

Key Takeaways & The Double-Counting Trap

• Robust Baseline: The median and mean inspection score across SF sits comfortably at 91.3 / 100.
• Concentrated Risk: Severe hygiene degradation is not systemic across the city—only 1.86% (275 sessions) scored in the critical 'Poor (<70)' bracket.
• The Double-Counting Pitfall: In raw data, each row represents 1 violation. Restorans with multiple violations appeared up to 8 times, artificially depressing raw averages to 86.2.
• Session-Level Aggregation: Aggregating to 1 row per inspection session recovered true operational reality (91.3), preventing false regulatory alarms.

FINDING 2: VIOLATION ANALYSIS

Pareto Principle: Top 10 Infractions Account for 58% of Total

Operational Hygiene vs Critical Hazards

- Physical Facility Issues Dominate: #1 most frequent violation is 'Unclean floors, walls, ceilings' (3,545 incidents), followed by handwashing sink accessibility.
- 86% Low/Moderate Risk: The overwhelming majority of recorded non-compliances are structural and sanitary maintenance issues, not active biological pathogens.
- Critical High-Risk Exception: 'High-risk food holding temperature' ranks #10 overall with 1,629 occurrences. This is the single largest biological food-poisoning hazard.
- Action Strategy: SFDPH checklist should decouple minor structural citations from urgent bacterial incubation risks.

FINDING 3: GEOGRAPHIC PROFILING

Geospatial Exposure: High-Density Dining Hubs Carry Elevated Risk

Statistical Thresholding & Risk Zones

- Small-Sample Distortion Avoided: Postal codes with <100 inspections (e.g. 94014, 94120) showed deceptive 50% rates due to sample sizes of 5-8 restaurants.
- Core Business District Vulnerability: Downtown corridors—94104 (Financial South, 22.9%) and 94122 (Sunset North, 20.5%)—exhibit statistically significant higher high-risk proportions.
- Why High-Density Corridors Suffer: Fast kitchen turnover, older infrastructure, and high dining traffic stress cold-storage equipment and pest barriers.
- Resource Reallocation: Moving 20% of inspector hours from low-risk suburbs to 94104, 94122, and 94133 maximizes public health impact.

FINDING 4: TEMPORAL SEASONALITY

Early Warning: +85% Surge in Vermin Infestations During Summer

Predictive Public Health Intervention

• Clear Seasonal Cyclicality: Rodent and insect infestations exhibit intense weather dependency, peaking in May (332 incidents) and June (353 incidents).
• Winter Lows: In contrast, December and January register only 162-190 incidents (a +85% climb from winter to summer).
• The Double Hazard: Summer temperature surges simultaneously degrade outdoor waste and stress refrigeration units, compounding pest attraction.
• Preventive Campaign Window: Rather than issuing fines in July, SFDPH should mandate pre-summer pest control audit notices across all districts in April.

FINDING 5: REMEDIATION IMPACT

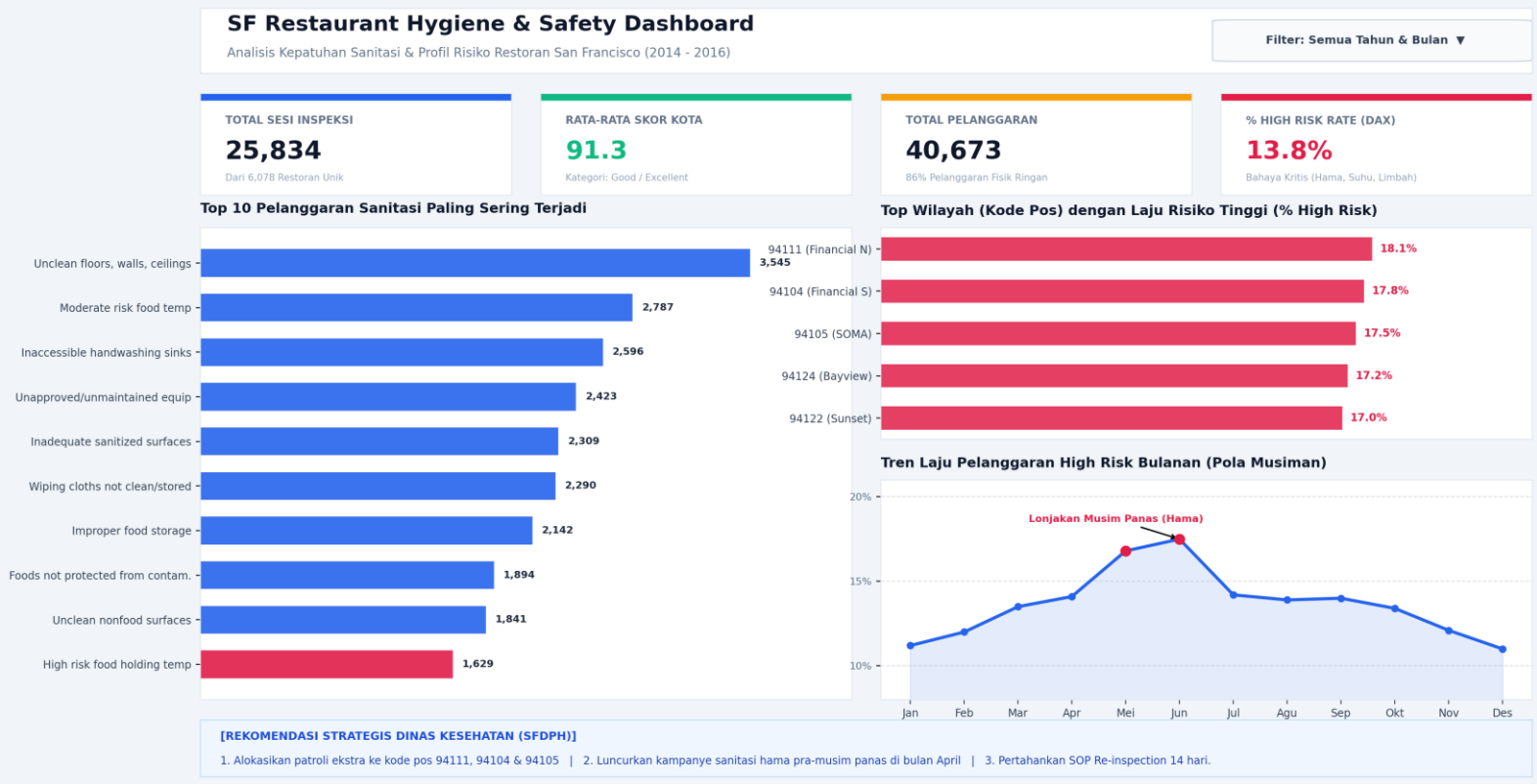
Enforcement Efficacy: Failing Venues Rebound by +8.57 Points

Measuring Regulatory Compliance ROI

• Follow-Up Analysis: Evaluated 730 restaurants that scored Poor/Substandard (<80) using SQL Window Functions (LAG) on subsequent re-inspections.
• Statistically Significant Recovery: Average initial score of 72.92 rose to 81.49 on follow-up—an average recovery of +8.57 points.
• 76% Success Rate: Over three-quarters of inspected venues successfully exited the danger zone upon mandatory re-inspection.
• Policy Takeaway: Routine follow-up inspections are not merely punitive; they serve as a powerful behavioral correction mechanism.

BI DELIVERABLE SHOWCASE

Executive Power BI Dashboard: Interactive Surveillance Center



Dashboard Highlights

- 16:9 HD Widescreen Executive Layout.
- 4 Synchronized KPI Cards with dynamic DAX calculations.
- Cross-Filtering Enabled: Selecting a district updates violations and monthly trends simultaneously.
- Color Hierarchy: Emerald (Safe), Amber (Attention), Crimson (Critical High-Risk).
- Executive micro-copy tailored for non-technical leadership.

DEPARTMENT NEXT STEPS

Strategic Action Plan: 3 High-Impact Recommendations for SFDPH

01

GEOGRAPHIC RESOURCE REALLOCATION —

Shift 20% Inspector Capacity to Downtown Hotspots

Reallocate inspector patrol shifts from low-density, compliant residential areas into high-risk commercial zones (94104, 94122, 94133). Introduce unannounced peak-hour visits during lunch/dinner rushes where food temperature violations peak.

02

PRE-SUMMER PEST ABATEMENT CAMPAIGN —

Mandatory April Notification & Waste Audits

Launch a targeted preventive pest abatement campaign every April, one month prior to the historical May-June infestation surge (+85%). Partner with municipal waste management to inspect outdoor dumpster seals and grease traps.

03

STANDARDIZED 14-DAY RE-INSPECTION PROTOCOL —

Institutionalize Rapid Regulatory Follow-Up

Maintain and formalize the mandatory 14-day re-inspection requirement for any establishment scoring below 80. Data proves that 76% of venues achieve compliance on re-check (+8.57 point gain), preventing repeat contamination.

Thank You!

Questions & Discussion

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